

A Complete Beginner's Course

Fundamentals · Key Concepts · Practical Examples

10 Topics

Real Examples

10 Visuals

1 · What is a Time Series?

2 · Components of Time Series

3 · Trends & Patterns

4 · Seasonality

5 · Stationarity

6 · Autocorrelation (ACF & PACF)

7 · Smoothing Techniques

8 · Forecasting Basics

9 · ARIMA Models

10 · Anomaly Detection

Table of Contents

Chapter 1: What is a Time Series?

Introduction and real-world examples

Chapter 2: Components of Time Series

Trend, Seasonality, Cyclicity, Noise

Chapter 3: Trends & Patterns

Types of trends and how to identify them

Chapter 4: Seasonality

Periodic patterns and their importance

Chapter 5: Stationarity

Why it matters and how to achieve it

Chapter 6: ACF & PACF

Tools for model identification

Chapter 7: Smoothing Techniques

Moving averages and exponential smoothing

Chapter 8: Forecasting Basics

Methods and confidence intervals

Chapter 9: ARIMA Models

The workhorse of time series forecasting

Chapter 10: Anomaly Detection

Spotting outliers and unusual patterns

Chapter 1

What is a Time Series?

A time series is any sequence of data points collected or recorded at successive points in time — typically at uniform intervals. Understanding time series is the first step toward making sense of temporal data.

Definition

A **time series** is an ordered sequence of values measured at equally-spaced intervals. Unlike cross-sectional data (a snapshot at one point in time), time series data captures *how* something evolves over time. The key feature is that **the order of observations matters** — shuffling the data destroys important information.

Key Insight: In time series analysis, time is always the independent variable. The goal is to understand how a measured quantity changes with time, and to use that understanding to make future predictions.

Real-World Examples

Time series data appears in virtually every field. Here are some common examples:

Finance	Stock prices, exchange rates, trading volumes, portfolio returns
Economics	GDP growth, inflation rates, unemployment figures, consumer spending
Healthcare	Heart rate measurements, patient vitals, disease case counts (e.g. COVID-19)
Meteorology	Daily temperature, rainfall, wind speed, humidity
Engineering	Sensor readings, machine vibrations, network traffic, server load
Retail	Daily sales, inventory levels, foot traffic, online orders

Time Series in the Real World

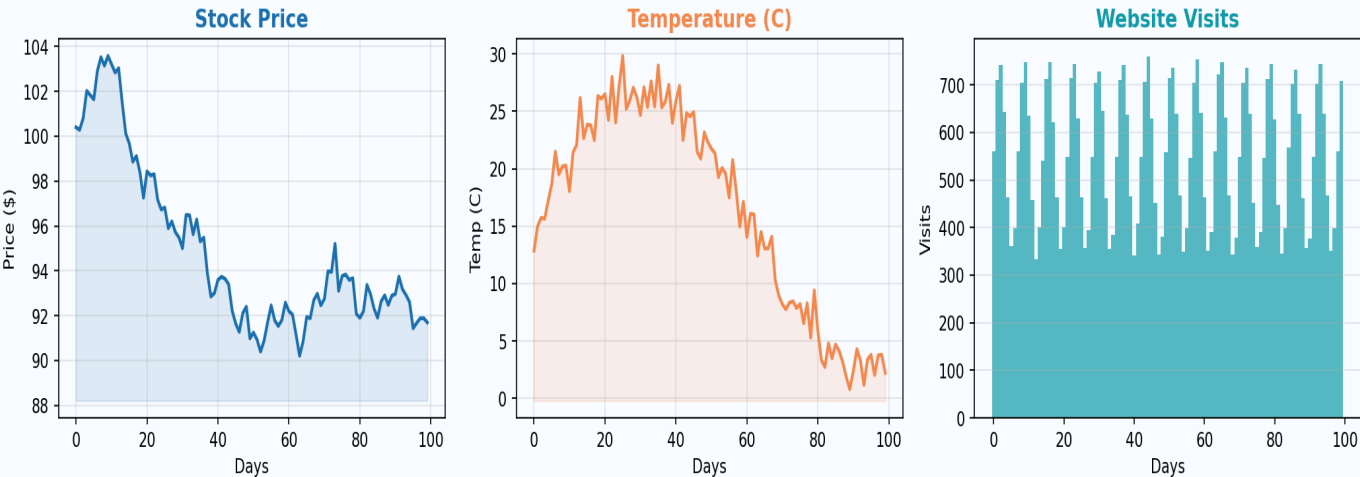


Figure 1.1 — Three real-world time series: stock prices (left), temperature fluctuations (centre), and website traffic (right). Each captures a different kind of temporal pattern.

Why Analyze Time Series?

Time series analysis serves three primary goals:

- **Description:** Understanding the historical behaviour of a variable — identifying trends, cycles, and unusual events.
- **Explanation:** Uncovering relationships between variables evolving over time, such as how advertising spend affects future sales.
- **Forecasting:** Predicting future values based on observed patterns — the most commercially valuable application of time series methods.

Remember: Good time series analysis begins with a simple plot. Always visualise your data before applying any statistical techniques!

Chapter 2

Components of a Time Series

Any time series can be understood as a combination of four fundamental components. Decomposing a series into these components is one of the most powerful ways to understand its structure.

The Four Components

Most time series are composed of the following elements, which can combine additively ($Y = T + S + C + R$) or multiplicatively ($Y = T \times S \times C \times R$):

Trend (T)	The long-term, smooth directional movement of the series — upward, downward, or flat. Example: smartphone adoption growing steadily over a decade.
Seasonality (S)	Regular, predictable fluctuations that repeat at known, fixed intervals (daily, weekly, yearly). Example: ice cream sales peaking every summer.
Cyclical ity (C)	Irregular wave-like fluctuations with no fixed period, usually lasting 2–10 years. Example: business cycles in the economy.
Residual / Noise (R)	The random, unexplained variation that remains after all other components are removed. Example: sudden news-driven stock price swings.

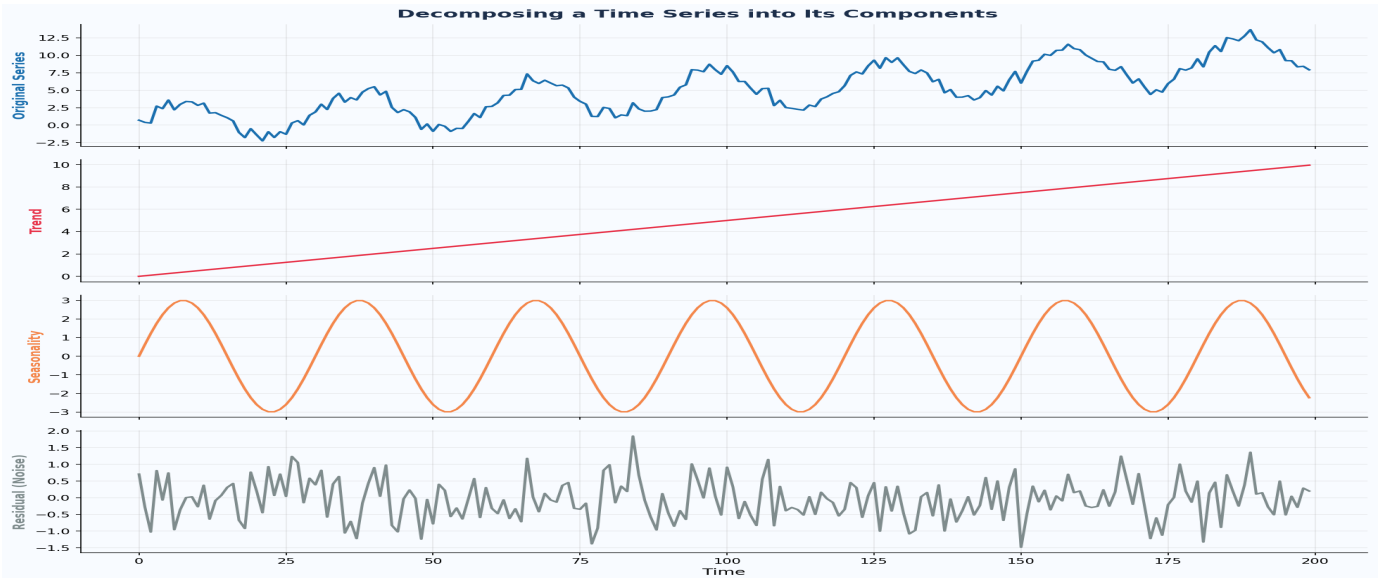


Figure 2.1 — A synthetic time series (top) decomposed into its trend, seasonal, and residual components. Note how summing the three lower panels recreates the top panel.

Additive vs Multiplicative Models

Choosing the right decomposition model is important:

--	--	--

Additive	$Y = T + S + C + R$	Seasonal fluctuations stay constant in magnitude over time
Multiplicative	$Y = T \times S \times C \times R$	Seasonal fluctuations grow proportionally with the level of the series

Chapter 3

Trends & Patterns

A trend describes the overall, long-term direction of a time series. Recognising the type of trend helps us choose the right model and interpretation.

Types of Trends

Linear Trend	Values increase or decrease at a <i>constant rate</i> . Modelled with a straight line: $Y = a + b \cdot t$. Example: steady population growth.
Exponential Trend	Values grow (or decay) at an <i>accelerating rate</i> . Common in technology adoption and compound interest.
Downward Trend	Values systematically decrease over time. Example: cost of solar panels declining year over year.
Flat / No Trend	No systematic direction; the series oscillates around a constant mean. This is one property of a stationary series.
Structural Break	An abrupt, permanent change in the trend direction, often caused by policy changes, crises, or technological disruption.

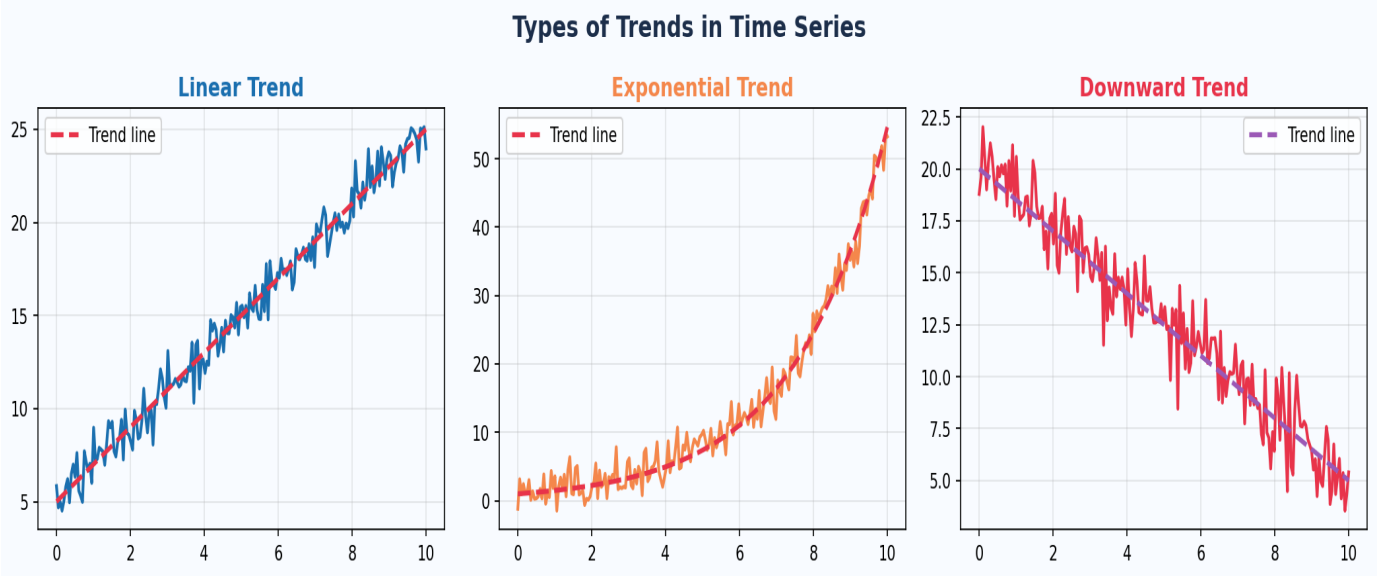


Figure 3.1 — Linear trend (left), exponential trend (centre), and downward trend (right). The dashed red line shows the fitted trend component in each case.

How to Identify Trends

- **Visual inspection:** Plot the data. A clear upward or downward movement is often visible.
- **Moving average:** Apply a moving average to smooth out short-term fluctuations and reveal the trend.
- **Regression:** Fit a linear (or polynomial) regression line to the time index.
- **Hodrick-Prescott filter:** A statistical technique used especially in economics to separate trend from cycle.

Practical Tip: Trends are your friend. Instead of trying to eliminate them immediately, first understand what drives them — this context often reveals the most important insights in your data.

Chapter 4

Seasonality

Seasonality refers to regular, periodic fluctuations that occur at fixed intervals. Understanding seasonality is crucial for accurate forecasting and business planning.

What is Seasonality?

Seasonal patterns arise because of external factors that repeat predictably — the calendar, human behaviour, or natural cycles. The period can be daily (e.g. hourly electricity demand), weekly (e.g. restaurant foot traffic), monthly (e.g. payroll cycles), or annual (e.g. retail holiday sales).

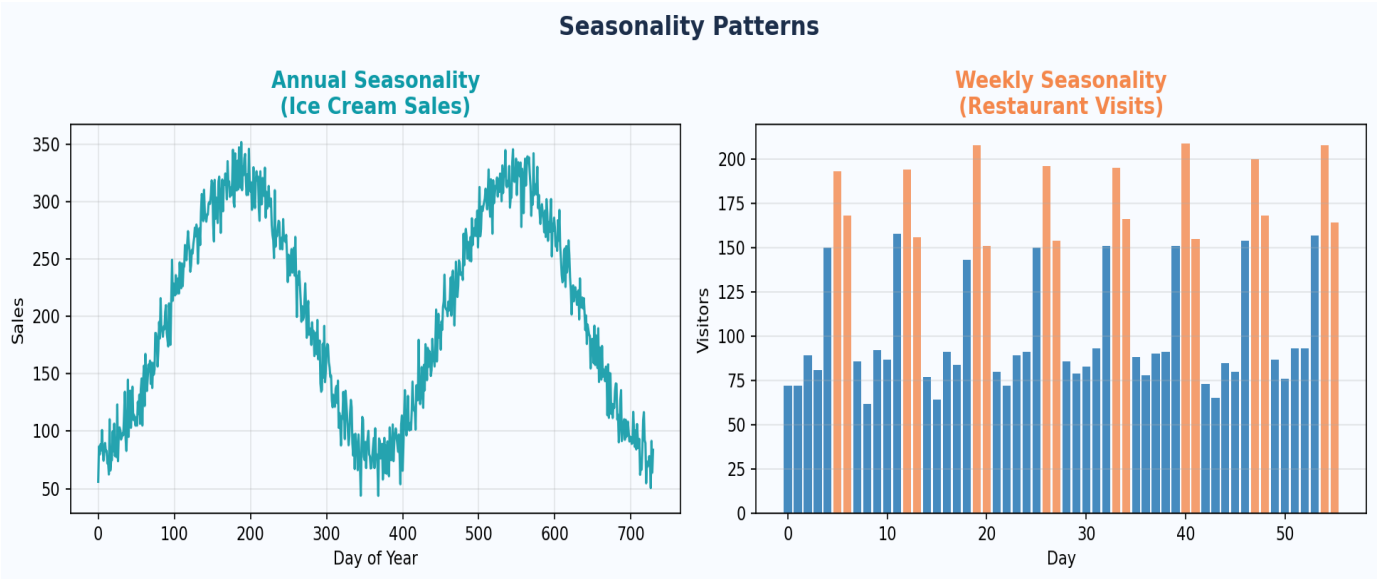


Figure 4.1 — Annual seasonality in ice cream sales (left) and weekly seasonality in restaurant visits (right). Orange bars highlight weekend peaks.

Seasonal vs Cyclical Variation

Property	Seasonality	Cyclicity
Period length	Fixed and known	Variable, unknown
Duration	Short (days to 1 year)	Long (2–10 years)
Cause	Calendar, human habits	Economic conditions
Predictability	High	Lower

Seasonal Adjustment

Seasonal adjustment (also called *deseasonalisation*) removes the seasonal component so we can see the underlying trend more clearly. Government statistical agencies routinely publish seasonally-adjusted economic data to help compare monthly figures fairly.

- **Classical decomposition:** Estimate and subtract the average seasonal factor at each period.
- **X-13ARIMA-SEATS:** The industry-standard method used by the US Census Bureau and Eurostat.

- **STL decomposition:** A robust, flexible method using LOESS smoothing — great for messy real-world data.

Chapter 5

Stationarity

Stationarity is one of the most important concepts in time series analysis. Most classical models assume the series is stationary — if it isn't, we need to transform it first.

What is Stationarity?

A time series is **strictly stationary** if its statistical properties (mean, variance, autocorrelation) do not change over time. In practice we usually only require **weak (covariance) stationarity**:

- The **mean** is constant over time.
- The **variance** is constant over time (no heteroscedasticity).
- The **autocorrelation** depends only on the lag, not on the specific time point.

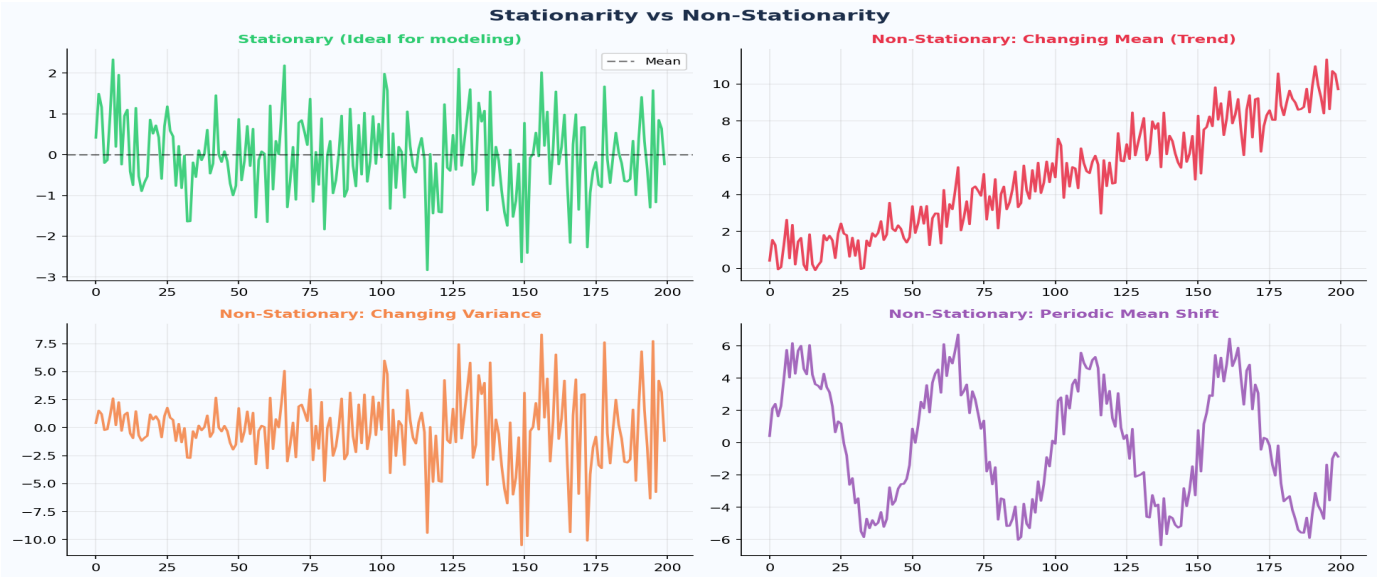


Figure 5.1 — A stationary series (top-left) compared with three non-stationary series: changing mean due to trend (top-right), changing variance (bottom-left), and periodic mean shifts (bottom-right). Only the green series is suitable for direct modelling.

Testing for Stationarity

ADF Test	Augmented Dickey-Fuller: tests whether a unit root is present. A significant p-value (< 0.05) means the series IS stationary.
KPSS Test	Kwiatkowski-Phillips-Schmidt-Shin: the null hypothesis is stationarity. Used alongside ADF for confirmation.
PP Test	Phillips-Perron: similar to ADF but robust to heteroscedasticity in the errors.
Visual ACF	A slowly-decaying ACF plot is a strong visual signal of non-stationarity.

Making a Series Stationary

- **Differencing:** Subtract consecutive values ($Y_t - Y_{t-1}$). First-order differencing removes a linear trend; second-order removes a quadratic trend.

- **Log transformation:** Stabilises variance in series with exponential growth (e.g. financial returns).
- **Detrending:** Fit and remove a regression trend.
- **Seasonal differencing:** Subtract the value from the same period in the previous cycle to remove seasonality: $Y_t - Y_{t-m}$ where m is the seasonal period.

Rule of Thumb: If the ADF test gives $p < 0.05$, your series is likely stationary. If not, try first-order differencing and test again. Most real-world series become stationary after one or two rounds of differencing.

Chapter 6

Autocorrelation: ACF & PACF

Autocorrelation measures how related a time series is to its own past values. The ACF and PACF plots are essential diagnostic tools for identifying the right model.

What is Autocorrelation?

The **autocorrelation** at lag k measures the linear relationship between a time series and a version of itself shifted k periods into the past. If today's temperature correlates with yesterday's, we have positive autocorrelation at lag 1.

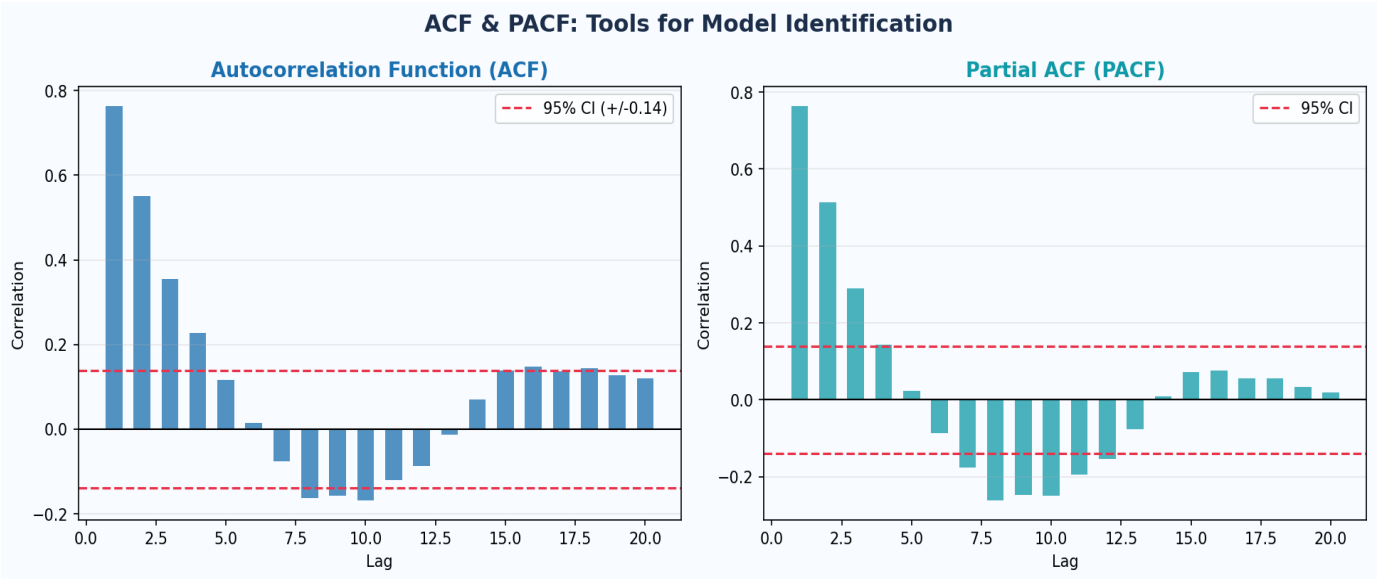


Figure 6.1 — ACF (left) showing the total correlation at each lag for an AR(1) process ($\rho=0.8$). The gradually decaying pattern is a hallmark of autoregressive behaviour. PACF (right) shows only the direct effect after removing intermediate lag effects.

ACF vs PACF

ACF (Autocorrelation Function)	Measures the correlation between y_t and y_{t-k} for ALL paths. Includes indirect effects.
PACF (Partial ACF)	Measures the correlation between y_t and y_{t-k} AFTER controlling for intermediate lags. Isolates the direct relationship.

Using ACF & PACF for Model Identification

Process	ACF Shape	PACF Shape
AR(p)	Decays gradually (exponential or damped sine)	Cuts off after lag p
MA(q)	Cuts off after lag q	Decays gradually

ARMA(p,q)	Decays gradually	Decays gradually
White Noise	No significant lags	No significant lags

Chapter 7

Smoothing Techniques

Smoothing methods reduce noise and help reveal the underlying signal in a time series. They are also used directly for short-term forecasting.

Simple Moving Average (SMA)

The SMA replaces each value with the average of a window of surrounding values. A wider window produces a smoother curve but lags behind turning points.

Formula: $SMA_t = (Y_{\{t\}} + Y_{\{t-1\}} + \dots + Y_{\{t-n+1\}}) / n$

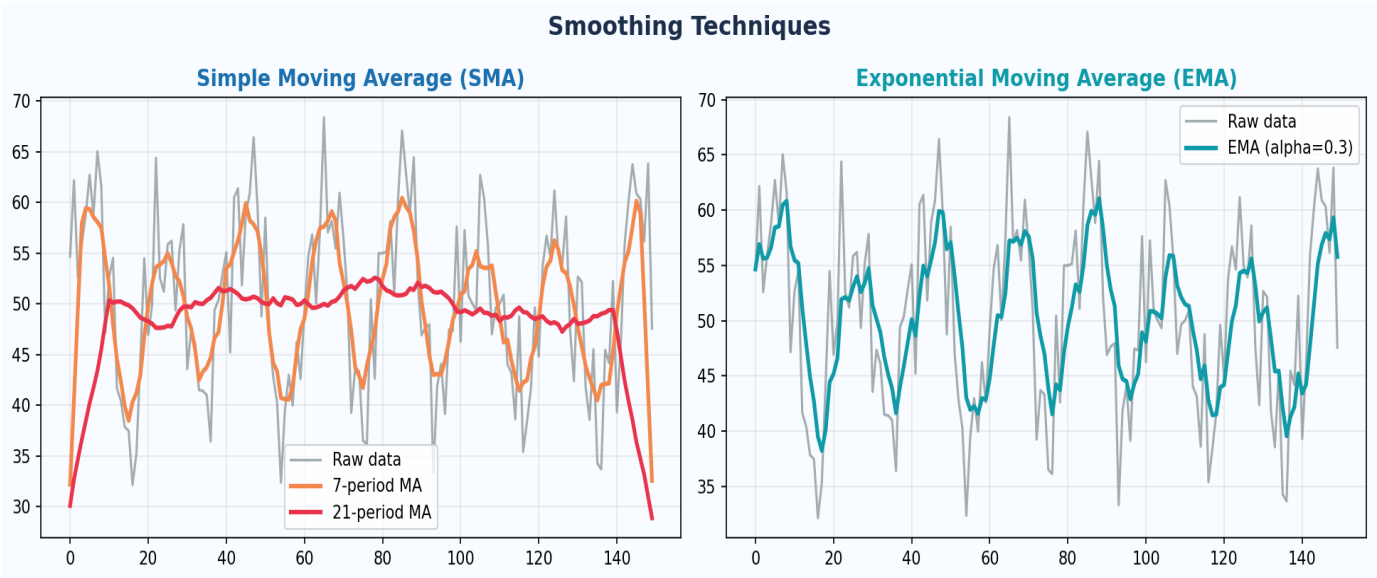


Figure 7.1 — Simple Moving Average (left): a 7-period MA (orange) and 21-period MA (red) applied to noisy data (grey). Exponential Moving Average (right): the EMA (teal) gives more weight to recent observations, reacting faster to changes.

Exponential Smoothing

Exponential smoothing gives *exponentially decreasing weights* to past observations, so recent data has more influence than older data. The smoothing parameter alpha ($0 < \alpha < 1$) controls the trade-off:

$EMA_t = \alpha * Y_t + (1 - \alpha) * EMA_{\{t-1\}}$

- **alpha close to 1:** reacts quickly to changes, but is noisy.
- **alpha close to 0:** very smooth, but slow to respond to genuine changes.

Advanced Smoothing Methods

Double Exponential (Holt's)	Adds a trend component — good for series with a trend but no seasonality.
Triple Exponential (Holt-Winters')	Adds both trend and seasonal components. One of the most widely used practical forecasting methods. Available in both additive and multiplicative variants.

LOESS / LOWESS	Local regression smoothing — very flexible, handles non-linear trends well. Used in STL decomposition.
----------------	---

Chapter 8

Forecasting Basics

Forecasting is the ultimate goal of time series analysis: using the past to predict the future. Understanding forecast uncertainty is just as important as the point forecast itself.

Types of Forecasting Methods

Naive Forecast	Tomorrow = Today. The simplest possible baseline — surprisingly hard to beat in highly volatile series.
Seasonal Naive	Tomorrow = Same period last season. Great baseline for seasonal data.
Exponential Smoothing	Weighted average of past values — good for data with trend and seasonality.
ARIMA	Statistical model capturing autocorrelation structure. The industry standard.
Machine Learning	Gradient boosting, LSTMs, and Transformer-based models for complex patterns.
Ensemble	Combining multiple forecasts (weighted average) often outperforms any single model.

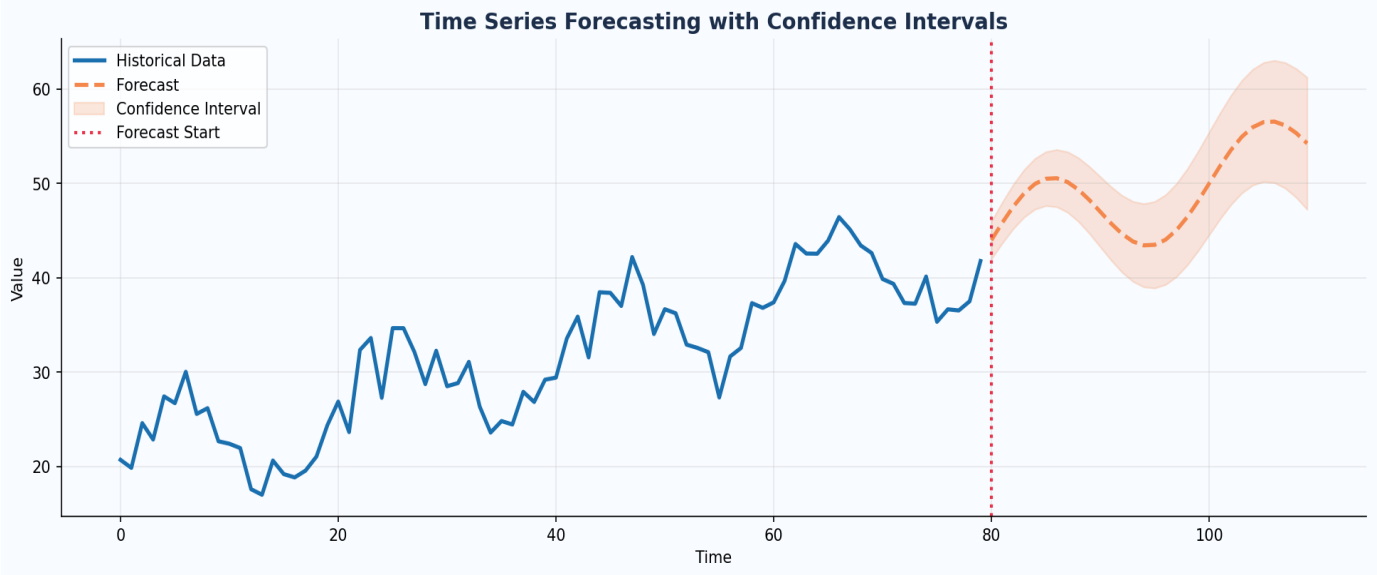


Figure 8.1 — A time series forecast showing historical data (blue), the predicted future values (orange dashed), and a widening confidence interval (shaded region). The interval grows wider further into the future, reflecting increasing uncertainty.

Forecast Accuracy Metrics

Always evaluate your model on a hold-out test set — data the model has never seen:

MAE	Mean Absolute Error — average magnitude of errors. Easy to interpret.
-----	---

RMSE	Root Mean Squared Error — penalises large errors more than MAE. Standard benchmark.
MAPE	Mean Absolute Percentage Error — scale-independent, great for comparing across series.
MASE	Mean Absolute Scaled Error — compares model to the naive baseline. $MASE < 1$ means you beat naive.

Golden Rule of Forecasting: Always compare your model to a naive baseline before claiming it is 'good'. A complex ARIMA model that barely beats a naive forecast may not be worth the added complexity!

Chapter 9

ARIMA Models

ARIMA (AutoRegressive Integrated Moving Average) is the most widely used framework for time series forecasting. It combines three powerful components into one flexible model.

The Three Components of ARIMA

AR — AutoRegressive (p)	The model regresses on its own past values. Parameter p is the number of past values (lags) used.
I — Integrated (d)	The degree of differencing applied to make the series stationary. d=1 means first-order differencing; d=0 means no differencing needed.
MA — Moving Average (q)	The model uses past forecast errors as inputs. Parameter q is the number of lagged error terms.

ARIMA(p,d,q) — The Full Model

The combined ARIMA(p,d,q) model is written as:

$$y'_t = c + \phi_1 y'_{t-1} + \dots + \phi_p y'_{t-p} + \theta_1 e_{t-1} + \dots + \theta_q e_{t-q} + e_t$$

where y'_t is the differenced series (applied d times) and e_t is white noise.

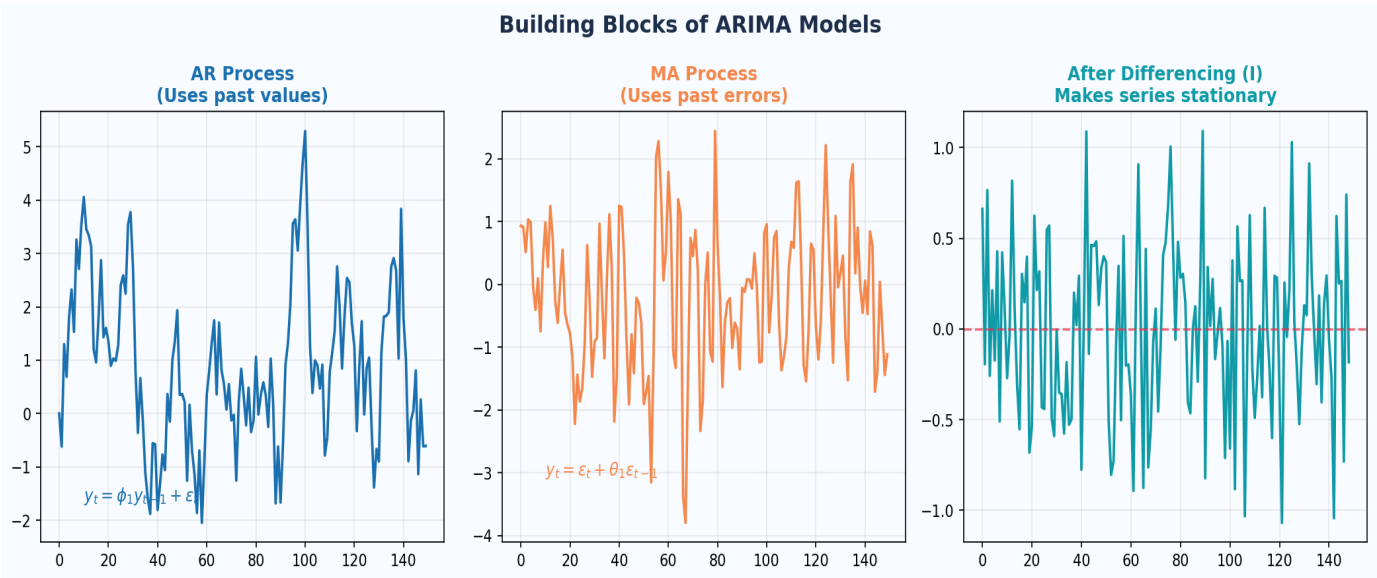


Figure 9.1 — The building blocks of ARIMA: an AR(1) process with $\phi=0.75$ (left), an MA(1) process with $\theta=0.7$ (centre), and first-order differencing to remove a random walk trend (right). Combining these gives a full ARIMA model.

Choosing ARIMA Parameters

- **Step 1:** Check stationarity (ADF/KPSS test). Apply differencing if needed — this gives d.
- **Step 2:** Plot ACF to find q (MA order): the lag where ACF cuts off.
- **Step 3:** Plot PACF to find p (AR order): the lag where PACF cuts off.
- **Step 4:** Fit the model and check residuals — they should look like white noise.

- **Step 5:** Use information criteria (AIC, BIC) to compare competing models automatically.

SARIMA: For seasonal data, extend ARIMA to SARIMA(p,d,q)(P,D,Q)_m, which adds seasonal AR, differencing, and MA terms at the seasonal period m. For monthly data with annual seasonality, m=12.

Chapter 10

Anomaly Detection

Anomalies (also called outliers or changepoints) are observations that deviate significantly from the expected pattern. Detecting them quickly is critical in fields like fraud detection, industrial monitoring, and cybersecurity.

Types of Anomalies

Point Anomaly	A single observation that is far from the rest. Most common type. Example: a sudden spike in server response time.
Contextual Anomaly	An observation that is anomalous in a specific context but not globally. Example: 30°C temperature in winter (anomalous) vs. summer (normal).
Collective Anomaly	A sequence of observations that is anomalous together, even if individual points seem normal. Example: a slow memory leak in a server.
Changepoint	A permanent shift in the statistical properties of the series. Example: a step change in a sensor reading indicating equipment failure.

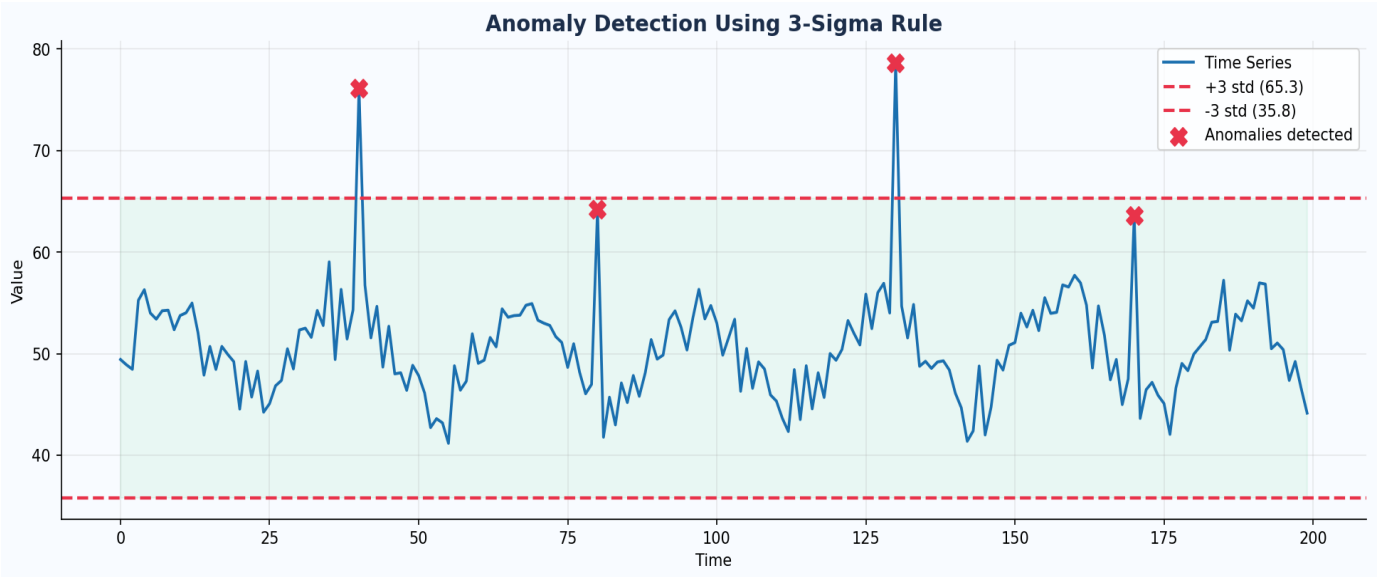


Figure 10.1 — Anomaly detection using the 3-sigma rule. Data points falling outside the 3 standard deviation bounds (dashed red lines) are flagged as anomalies (red X markers). The green shaded region is the 'normal' zone.

Common Detection Methods

3-Sigma Rule	Simple and fast. Flag any point more than 3 standard deviations from the mean. Works best for stationary, normally distributed data.
IQR Method	Flag points below $Q1 - 1.5 \cdot IQR$ or above $Q3 + 1.5 \cdot IQR$. More robust to non-normality.

Moving Average Control	Compare each point to the local moving average. Detects sudden changes even in trending or seasonal data.
Isolation Forest	Machine learning method: anomalies are 'easier to isolate' in the feature space. Handles multivariate and complex patterns well.
CUSUM / EWMA Charts	Sequential monitoring methods widely used in manufacturing quality control. Very sensitive to small persistent shifts.

Important: Not all anomalies are bad! Some represent genuine rare events worth investigating (e.g. a viral marketing campaign causing a sales spike). Always combine statistical detection with domain knowledge before acting on an anomaly.

Quick Reference: Key Concepts

Time Series	Ordered sequence of values recorded over time
Trend	Long-term directional movement (up, down, or flat)
Seasonality	Regular, repeating pattern at a fixed period
Cyclicalitiy	Irregular wave-like fluctuations over years
Stationarity	Statistical properties (mean, variance) constant over time
Differencing	Subtracting consecutive values to remove a trend
ACF	Correlation of a series with its lagged self
PACF	Direct correlation with a lag, controlling for intermediate lags
Moving Average	Rolling average over a window of past values
Exponential Smoothing	Weighted average giving more weight to recent values
ARIMA(p,d,q)	AutoRegressive Integrated Moving Average model
AR(p)	Model using p past values of the series
MA(q)	Model using q past forecast errors
SARIMA	Seasonal extension of ARIMA
ADF Test	Test for stationarity — $p < 0.05$ means stationary
MAE / RMSE / MAPE	Forecast accuracy metrics
Anomaly	Observation deviating significantly from expected pattern
Changepoint	Permanent shift in the series' statistical properties
Decomposition	Splitting a series into Trend + Seasonal + Residual
White Noise	Purely random series with no structure — ideal residuals

Next Steps in Your Time Series Journey

- **Python libraries to explore:** statsmodels, pmdarima, prophet (Meta), sktime, darts
- **R packages:** forecast, tseries, fable, tsibble
- **Recommended reading:** 'Forecasting: Principles and Practice' by Hyndman & Athanasopoulos (free online at otexts.com/fpp3)
- **Practice datasets:** M4 Competition, Kaggle time series competitions, Federal Reserve FRED database

- **Advanced topics:** Vector Autoregression (VAR), state-space models, Gaussian Processes, Temporal Fusion Transformers, N-BEATS

Congratulations on completing this course! Time series analysis is a vast field, but with the fundamentals covered here you are equipped to explore real datasets, build forecasting models, and detect anomalies with confidence. The best way to consolidate your knowledge is to apply these techniques to data you care about.

